the CISO, the members of the Security Working Group (and their designees), and Commission observers of the Security Working Group.

The Plan Processor would incur costs to develop policies and procedures governing the review of applications for exceptions to the SAW use requirement. The Commission preliminarily estimates that the Plan Processor will incur labor costs of $56,000[689] to develop these policies and procedures, and annual ongoing costs of $31,700[690] to maintain and update these policies and procedures.

The Plan Processor will incur costs to review exception applications.[691] Each initial exception application would cause the Plan Processor to incur one-time labor costs of approximately $91,760.[692] Review of materials for continuation of exceptions would cause the Plan Processor to incur the same review costs annually.

The Plan Processor will incur costs to notify the Operating Committee that each Excepted Environment is compliant with the detailed design specifications that Participants provide as part of their application materials for an exception.[693] The Commission preliminarily estimates that the Plan Processor will incur $18,550[694] in labor costs to evaluate each Excepted Environment

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689 See supra note 523.

690 See supra note 524.

691 See supra Part III.D.3.d.ii.

692 See supra Part III.D.3.d.ii. The PRA estimates that the Plan Processor would incur $91,760 in labor costs to review each application. In this analysis, the Commission assumes all nine Participant Groups would apply for exceptions. (9 Participant Groups x $91,760 per application) = $825,840.

693 Id.

694 See supra note 531. The PRA estimates that the Plan Processor would incur $18,550 in labor costs to validate each Excepted Environments. In this analysis, the Commission assumes all nine Participant Groups would apply for exceptions. (9 Participant Groups x $18,550 per validation) = $166,950.

and notify the Operating Committee. Should the Plan Processor need to notify a Participant Group of an identified non-compliance with the detailed design specifications, additional costs would be incurred.[695]

The Plan Processor will incur costs to monitor the Excepted Environments in accordance with the detailed design specifications and notify the Participant of any identified non-compliance. The Commission preliminarily estimates the Plan Processor will incur annual ongoing costs of $302,600[696] to perform these tasks.

The proposed amendments require that each Participant using a non-SAW environment simultaneously notify the Plan Processor, the members of the Security Working Group (and their designees), and Commission observers of the Security Working Group of any material changes to its security controls for the non-SAW environment. The Commission cannot predict how many such changes would occur because the Commission does not know how often each Participant Group would make changes to its Excepted Environment that would necessitate material changes to its security controls, but for each such instance, the Commission preliminarily estimates the notifying Participant Group would incur labor costs of approximately $5,200.[697]

The Commission recognizes that by providing an exception procedure to the requirement that Participants employ the user-defined direct query and bulk extract tools to access CAT Data within SAWs, variability across environments from where CAT Data is accessed and analyzed

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695 Id.

696 See supra note 534

697 Labor costs include 10 hour of Senior Systems Analyst labor, 3 hours by a compliance attorney, and 2 hours by the CISO. For the CISO, hourly rate calculations use the hourly rate for a Chief Compliance Officer. (10 hours x $291/hour + 3 hours x $374/hour + 2 hours x $543/hour) = $5,118.

will necessarily increase. The amendments will provide for a level of security in Excepted Environments that will be similar but not identical to security within SAWs because Excepted Environments may implement security controls, policies, and procedures differently than SAWs. The Commission preliminarily believes the risk of individual Excepted Environments being less secure than SAWs is mitigated by the review process of applications for exceptions and Plan Processor verification and monitoring steps required by the proposed amendments.

4. OTQT and Logging

The CAT NMS Plan does not limit the amount of CAT Data a regulator can extract or download through the online targeted query tool ("OTQT"); the CAT NMS Plan only states that the Plan Processor must define the maximum number of records that can be viewed in the OTQT as well as the maximum number of records that can be downloaded.[698]

The proposed amendments would remove the ability of the Plan Processor to define the maximum number of records that can be downloaded via the OTQT, and instead limit the maximum number of records that can be downloaded via the OTQT to no more than 200,000 records per query request.[699] The Plan does not explicitly prevent use of the OTQT to download significant quantities of CAT Data, although the OTQT does not provide access to all fields in transactional CAT Data that are available through the user defined direct query tool, ("UDDQ"). Because the Plan does not currently distinguish between what types of analytic environments (SAWs versus Excepted Environments) may access particular tools (i.e., OTQT versus UDDQ), this may not be a significant security distinction under the Plan because downloading such data through the OTQT would be merely less efficient than doing so with other data extraction tools if

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698 See supra Part II.D.

699 See supra Part II.D.

either approach were available in a given analytic environment. However, with the proposed amendments' provisions that restrict the use of the UDDQ and bulk extract methods to Plan Processor provided SAWs and Excepted Environments, some regulatory users may be incentivized to use a succession of queries to download larger samples of CAT Data using the OTQT to avoid the need to work within the SAWs or Excepted Environments.

The Commission preliminarily believes that by limiting the number of records of CAT Data that can be extracted from the OTQT, the proposed amendments are likely to result in more regulatory analysis of CAT Data being performed within the security perimeter established by the CISP of the Plan Processor because regulatory activities that require extraction of more than 200,000 records would need to be performed using the UDDQ or by bulk extraction, activities that would be limited to Plan Processer provided SAWs or Excepted Environments under the proposed amendments. The Commission preliminarily believes that this is likely to reduce the attack surface of CAT by reducing the magnitude of CAT Data accessed outside of these potentially more secure environments. The Commission recognizes, however, that limiting the use of the OTQT to queries that extract fewer than 200,000 records may also reduce regulatory use of CAT Data to the extent that a regulatory user may not have the technical skills that would be required to use other access methods.[700]

The proposed amendments extend the information in log files that the Participants are required under the Plan to submit to the Operating Committee monthly, specifically, by defining the term "delivery of results" and requiring the logging of access and extraction of CAT Data.[701]

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700 The Commission preliminarily believes that access to CAT Data through the UDDQ would require greater technical skills on the part of the user such as knowledge of a structured query language and an understanding of structured databases.

701 See supra Part III.D.4.

The Commission estimates that the Plan Processor will incur one-time labor costs of $87,960[702] to make the initial necessary programming and systems changes to log delivery of results of queries of CAT Data and the access and extraction of CAT Data. In addition, the Plan Processor would incur an annual ongoing expense of $5,100[703] to generate and provide the additional information in monthly reports required by the proposed amendments. The Commission preliminarily estimates that the Participants would incur ongoing annual labor costs of $970,200[704] for the Operating Committee to review the additional information in the monthly reports. Further, the requirement that limits the number of records that can be extracted through use of the OTQT may make it impossible for some regulatory functions that are required only situationally (such as ad hoc queries to investigate trading by a single trader in all symbols or by multiple traders in a single symbol) to be performed outside the SAW (or Excepted Environments). This restriction may cause some Participants to establish SAWs, obtain an exception, or extend their use of RSAs for activities that are performed infrequently. This outcome may be more costly to these Participants than working less efficiently through the OTQT in ad hoc situations because it may be less costly to Participants to use the OTQT inefficiently than to make these alternative arrangements for only occasional use.

5. CAT Customer and Account Attributes

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702 See supra Part III.D.4.

703 See supra Part III.D.4.

704 Cost estimate assumes each Participant would annually incur 12 hours of Operating Committee Member labor and 108 hours of Compliance Manager labor. (12 hours x $381/hour + 108 hours x $317/hour) = $38,808 per Participant. Collectively, Participants would incur ($38,808 per Participant x 25 Participants) = $970,200.

As noted above, the Commission granted the Participants' PII Exemption Request to allow for an alternative approach to generating a Customer-ID and to allow for an alternative approach which would exempt the reporting of dates of birth and account numbers associated with retail customers who are natural persons.[705] This exemptive relief allows the Participants to implement an alternative approach to generating Customer-ID(s), subject to certain conditions set forth in the exemptive relief, but does not bar the Participants from implementing the Plan's original Customer-ID approach.

The baseline for customer and account information availability in CAT assumes the implementation of the alternative approach described in the PII Exemption Order and the creation of the CCID Subsystem. The exemptive relief includes certain conditions that also are included in the baseline for the proposed amendments.[706] First, the exemptive relief requires that the Participants "ensure the timeliness, accuracy, completeness, and integrity of interim value[s]" in the CCID Subsystem.[707] Second, the Participants must assess the overall performance and design of the CCID Alternative process and the CCID Subsystem as part of each annual Regular Written Assessment of the Plan Processor.

The Commission proposes to amend the CAT NMS Plan to: (1) delete the Industry Member reporting of ITINs/SSNs, dates of birth and account numbers for natural persons and require the reporting of year of birth; (2) establish a process for creating Customer-ID(s); (3) impose specific obligations on the Plan Processor that will support the revised reporting requirements and creation of Customer-ID(s); and (4) amend existing provisions of the CAT

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705 See supra Part II.E.

706 See PII Exemption Order, supra note 164, at 16157.

707 See id.

NMS Plan to reflect the new reporting requirements and process for creating Customer-ID(s), as further discussed below.[708] These provisions reflect the PII exemptive relief previously granted by the Commission.

The Commission preliminarily believes that the provisions of the proposed amendments discussed in this section largely reflect exemptive relief and current implementation specifications of the Participants, with the exception of the requirement that customer addresses reported to the CAIS have separate fields for street numbers and names. Because the specifications are still in development, the Commission preliminarily believes that the cost impact of this provision on Participants is likely to be de minimis. The Commission further preliminarily believes that CAT Reporters have not implemented an alternative street address specification and the costs to CAT Reporters to implement this change will be de minimis because the requirement does not require additional information to be reported.

The proposed amendments include provisions that by design, reduce certain options for future development of the Plan. For example, the Participants would not be able to decide at a later date to no longer use their exemptive relief and instead change the CAT implementation to conform to the Plan as it stands at that time. Although the Commission believes that the Participants would be unlikely to take such an approach in the future after incurring the costs to secure exemptive relief and implement alternative approaches required by such relief, it recognizes that the proposed amendments curtail that option to the Participants.

6. Customer Identifying Systems Workflow

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708 See supra Part II.E.

The Commission is proposing to amend the CAT NMS Plan to define the workflow for accessing Customer and Account Attributes, and to establish access restrictions.[709] Accordingly, the Commission proposes to amend the CAT NMS Plan to 1) specify how existing data security requirements apply to Customer and Account Attributes; 2) define the Customer Identifying Systems Workflow and the General Requirements for accessing Customer Identifying Systems; 3) establish general requirements that must be met by Regulatory Staff before accessing the Customer Identifying Systems, which access will be divided between two types of access - manual access and programmatic access; and 4) establish the specific requirements for each type of access to the Customer Identifying Systems. Some of these provisions would reflect the PII exemptive relief previously granted by the Commission, making the alternative approach described in the PII Exemption Order a requirement of the Plan. The Commission discusses potential benefits of the proposed new provisions of the Plan relative to the baseline below.

The proposed amendments would replace the term "PII" with "Customer and Account Attributes" and to reflect that Customer Identifying Systems, including CAIS, now contain the information that identifies a Customer; prohibit Customer and Account Attributes from being included in the result sets to queries of transactional CAT Data; and update requirements related to the PII access audit trail to reflect the CAIS approach. These requirements mirror requirements for access to customer information already contained in the Plan or the PII Exemptive Order.[710] The Commission preliminarily believes that these provisions may avoid inefficiencies in implementation to the extent that Participants might make investments in

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709 See supra Part II.F.

710 See supra Part II.F.1.

implementation activities that do not reflect the approach to customer information and account attributes outlined in the exemptive relief.

The proposed amendments include provisions that limit access to the Customer Identifying Systems to two types of access – manual and programmatic. The Commission preliminarily believes that this may improve the security of CAT Data by limiting access to CAIS data to two defined access methods. The Commission preliminarily believes that by doing so the likelihood that customer information might be compromised in a potential breach will be decreased. To the extent that a bad actor would be limited in his or her ability to access customer information in a manner other than these two access pathways, customer information within the CAT System should be more secure.

The proposed amendments include provisions that establish that access to Customer Identifying Systems are subject to certain restrictions, including requiring that authorization to use Programmatic CAIS Access or Programmatic CCID Subsystem Access be requested and approved by the Commission.[711] The Commission preliminarily believes that this authorization step may reduce the risk of inappropriate use of customer and account information by ensuring that programmatic access that can potentially return information about a large group of customers is only granted when an appropriate regulatory use exists. Further, the Commission preliminarily believes this requirement may reduce the amount of CAT Data exposed to regulators as they perform their duties because it may increase regulatory use of manual as opposed to programmatic access to the CCID Subsystem and CAIS when manual access is sufficient for a regulatory purpose.

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[711] See supra Part II.F.5.

The proposed amendments would establish programmatic access as a required element of the CAT NMS Plan.[712] The provision of programmatic access enables authorized Regulatory Staff to query the CAIS and CCID Subsystems to access information on multiple customers or accounts simultaneously.[713] The Commission recognizes that allowing programmatic access to CAIS and CCID data by authorized users potentially will allow Regulatory Staff to be exposed to a greater quantity of Customer and Account Attributes. To the extent that this exposure provides more opportunities for this data to be used inappropriately, this may reduce the confidentiality of CAIS and CCID data. However, the Commission preliminarily believes the Commission authorization step required before programmatic access can be exercised mitigates this risk because the application review process requires documentation establishing the regulatory purpose of the programmatic access, and provides for an approval process based on such access being generally consistent with specific standards that would justify such access.[714]

The Commission preliminarily estimates that the Plan Processor will incur labor costs of $620,200[715] to establish programmatic access to the CCID Subsystem and CAIS.

Under the proposed amendments, Participants that require programmatic access to the CAIS or CCID Subsystems would need to apply for authorization from the Commission.[716] The Commission cannot estimate how many Participants would need to apply for authorization, or

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712 See supra Part II.

713 See supra Part II.F.7.

714 See supra Part II.F.6.

715 The estimates assumes 640 hours each of labor by a Senior Database Administrator, a Senior Programmer and a Senior Business Analyst. (640 hours x $349/hour + 640 hours x $339/hour + 640 hours x $281/hour) = $620,160.

716 Id.